

## Substitution



**1** Find the value of:

**a**  $9 + m$  when  $m = 8$

**c**  $n - 2$  when  $n = 5$

**e**  $\frac{n}{7}$  when  $n = 14$

**g**  $u \times 5$  when  $u = 6$

**i**  $2x + 9$  when  $x = 4$

**k**  $x^2 - 4x$  when  $x = 3$

**m**  $\frac{3x}{4}$  when  $x = 8$

**o**  $(x + 2)(x - 1)$  when  $x = -3$

**b**  $14 + v$  when  $v = 15$

**d**  $n - 8$  when  $n = 44$

**f**  $\frac{u}{9}$  when  $u = 54$

**h**  $n \times 9$  when  $n = 8$

**j**  $y^2$  when  $y = -2$

**l**  $\frac{x + 10}{4}$  when  $x = 2$

**n**  $2(x + 7)$  when  $x = 4$

**p**  $4y^2 + 9$  when  $y = 2$

**2** Find the value of:

**a**  $m + n$  when  $m = 7$  and  $n = 12$

**c**  $x^2 - 4y$  when  $x = -2$  and  $y = 5$

**e**  $\frac{5xy}{9}$  when  $x = 3$  and  $y = 6$

**b**  $4x + 2y + 3$  when  $x = 5$  and  $y = 4$

**d**  $8a + 4b - 9$  when  $a = 2$  and  $b = 5$

**f**  $abc$  when  $a = 3, b = -2$  and  $c = -1$

**3** For each formula find the value of  $y$  for the given value of  $x$ :

**a**  $y = 6x - 9$ , when  $x = 12$ .

**c**  $y = 4 - 3x$ , when  $x = 7$ .

**b**  $y = \frac{x}{9} + 6$ , when  $x = 63$ .

**d**  $y = 14x + \frac{18}{5}$ , when  $x = \frac{1}{10}$ .

## Tables of values

**4** Consider the following formulae:

**a**  $q = p + 7$

**i** Complete the table:

|     |   |   |    |   |   |   |
|-----|---|---|----|---|---|---|
| $p$ | 1 | 2 | 3  | 4 | 5 | 6 |
| $q$ | 8 | 9 | 10 |   |   |   |

**ii** Find the value of  $q$  when  $p = 30$ .

**b**  $q = 3p$



**i** Complete the table:

**ii** Find the value of  $q$  when  $p = 20$ .

|     |   |   |   |   |   |   |
|-----|---|---|---|---|---|---|
| $p$ | 1 | 2 | 3 | 4 | 5 | 6 |
| $q$ | 3 | 6 | 9 |   |   |   |

**c**  $q = 6p + 4$

**i** Complete the table:

**ii** Find the value of  $q$  when  $p = 40$ .

|     |    |    |    |   |   |   |
|-----|----|----|----|---|---|---|
| $p$ | 1  | 2  | 3  | 4 | 5 | 6 |
| $q$ | 10 | 16 | 22 |   |   |   |

**d**  $n = m - 7$

**i** Complete the table:

**ii** Find the value of  $n$  when  $m = 20$ .

|     |    |    |    |    |    |    |
|-----|----|----|----|----|----|----|
| $m$ | 10 | 11 | 12 | 13 | 14 | 15 |
| $n$ | 3  | 4  | 5  |    |    |    |

**e**  $q = 5p - 3$ .

**i** Complete the table:

**ii** Find the value of  $q$  when  $p = 20$ .

|     |   |   |    |   |   |   |
|-----|---|---|----|---|---|---|
| $p$ | 1 | 2 | 3  | 4 | 5 | 6 |
| $q$ | 2 | 7 | 12 |   |   |   |

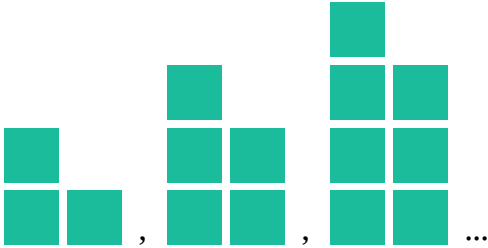
**f**  $q = p^2 + 5$ .

**i** Complete the table:

**ii** Find the value of  $q$  when  $p = 20$ .

|     |   |   |    |   |   |   |
|-----|---|---|----|---|---|---|
| $p$ | 1 | 2 | 3  | 4 | 5 | 6 |
| $q$ | 6 | 9 | 14 |   |   |   |

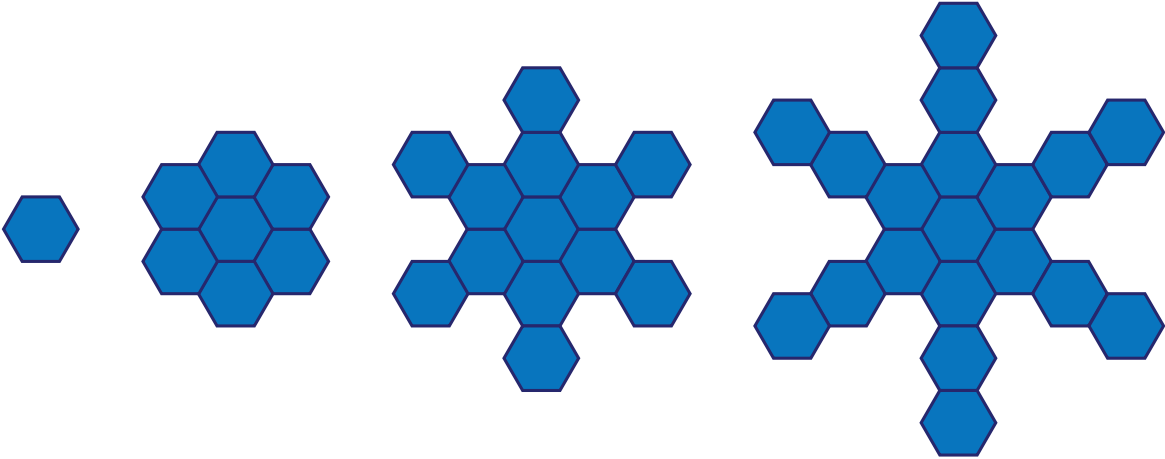
5 Vanessa is making the sequence to the right out of tiles. She creates a table comparing the sequence number of a shape to the number of tiles needed to make it:



| Sequence number ( $S$ ) | 1 | 2 | 3 | 4 | 5 | 6 |
|-------------------------|---|---|---|---|---|---|
| No. tiles ( $T$ )       | 3 | 5 | 7 |   |   |   |

- a Write an algebraic rule for the number of tiles,  $T$ , in terms of the sequence number,  $S$ .
- b Use your rule to complete the table of values.

6 James is making snowflakes out of hexagonal tiles:



James creates a table comparing the width of a snowflake to the number of tiles needed to make it:

| Width ( $W$ )     | 1 | 3 | 5  | 7  | 9 | 11 | 13 |
|-------------------|---|---|----|----|---|----|----|
| No. tiles ( $T$ ) | 1 | 7 | 13 | 19 |   |    |    |

- a Write an algebraic rule for the number of tiles,  $T$ , in terms of the snowflake's width,  $W$ .
- b Use your rule to complete the table of values.